Customer Behaviour Analysis

Project Documentation

1. Project Overview

Project Title: Customer Shopping Behavior Analysis

Goal: Understand purchasing patterns, customer demographics, and behavior drivers (discounts, subscriptions, shipping preferences, purchase frequency) to support decisions on marketing, retention, and inventory.

Tech Stack: Python (data cleaning & EDA) → PostgreSQL (storage & querying) → Power BI (dashboarding & visualization)

2. Data Summary

The dataset has 3,900 rows and 18 columns, one row per customer transaction. It covers basic information such as **Customer ID**, **Age** (18–70, avg 44), and **Gender** (mostly male). It also tracks what was bought — **Item Purchased**, **Category** (mostly Clothing), **Purchase Amount** (avg $59.76), **Size**, and **Color** — along with **Location** (50 states) and **Season**.

Customer habits are captured through **Review Rating** (avg 3.75, with 37 values missing), **Subscription Status** (about 27% are subscribers), and **Previous Purchases** (avg 25). It also includes **Shipping Type**, **Discount Applied**, **Promo Code Used**, **Payment Method**, and **Frequency of Purchases**. Overall, the data is clean, with Review Rating being the only column containing missing values (37 out of 3,900, ~1%).

3. Exploratory Data Analysis Using Python

3.1 Data Loading

- Imported the dataset CSV file using pandas.

3.2 Initial Data Exploration

- Used **df.info()** to check the structure of the dataset.
- Used **df.describe()** to generate summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	P
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	

```
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   str
 3   Item Purchased          3900 non-null   str
 4   Category                3900 non-null   str
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   str
 7   Size                    3900 non-null   str
 8   Color                   3900 non-null   str
 9   Season                  3900 non-null   str
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   str
 12  Shipping Type           3900 non-null   str
 13  Discount Applied        3900 non-null   str
 14  Promo Code Used         3900 non-null   str
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   str
 17  Frequency of Purchases  3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Applie
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	N
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	Na
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	Na
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	Na
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	Na
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	Na
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	Na
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	Na

3.3 Missing Data Handling

- Checked for null values using **df.isnull().sum()**.
- Filled missing values in the **Review Rating** column using the median rating of each product category.

Before

```
Customer ID                 0
Age                         0
Gender                      0
Item Purchased              0
Category                    0
Purchase Amount (USD)       0
Location                    0
Size                        0
Color                       0
Season                      0
Review Rating              37
Subscription Status         0
Shipping Type               0
Discount Applied            0
Promo Code Used             0
Previous Purchases          0
Payment Method              0
Frequency of Purchases      0
dtype: int64
```

After

```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

3.4 Column Standardization

- Renamed columns to snake_case for better readability and documentation consistency.

3.5 Feature Engineering

- **age_group** — created by binning customer ages.

- **purchase_frequency_days** — derived from the purchase date.

3.6 Data Consistency Check

- Verified that **discount_applied** and **promo_code_used** were redundant, then dropped **promo_code_used** to reduce data size.

3.7 Database Integration

- Connected the Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis Using SQL (Business Questions)

Performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender —** Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users —** Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90

Total rows: 839 Query complete 00:00:03.739

3. **Top 5 Products by Rating —** Found products with the highest average review ratings.

	product text	Avg Review rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Total rows: 5 Query complete 00:00:01.154

4. **Shipping Type Comparison —** Compared average purchase amounts between Standard and Express shipping.

	Avg Purchase Amount numeric	shipping_type text
1	58.46	Standard
2	60.48	Express

Total rows: 2 Query complete 00:00:00.727

5. **Subscribers vs. Non-Subscribers —** Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	average_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Total rows: 2 Query complete 00:00:00.525

6. **Discount-Dependent Products —** Identified the 5 products with the highest percentage of discounted purchases.

	product text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

Total rows: 5 Query complete 00:00:00.316

7. **Customer Segmentation —** Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customer bigint
1	Loyal	3116
2	New	83
3	Returning	701

Total rows: 3 Query complete 00:00:00.283

8. **Top 3 Products per Category —** Listed the most purchased products within each category.

	item_rank	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145

Total rows: 11 Query complete 00:00:00.756

9. **Repeat Buyers & Subscriptions —** Checked whether customers with more than 5 purchases are more likely to subscribe.

	repeat_buyers bigint	subscription_status text
1	2518	No
2	958	Yes

Total rows: 2 Query complete 00:00:00.475

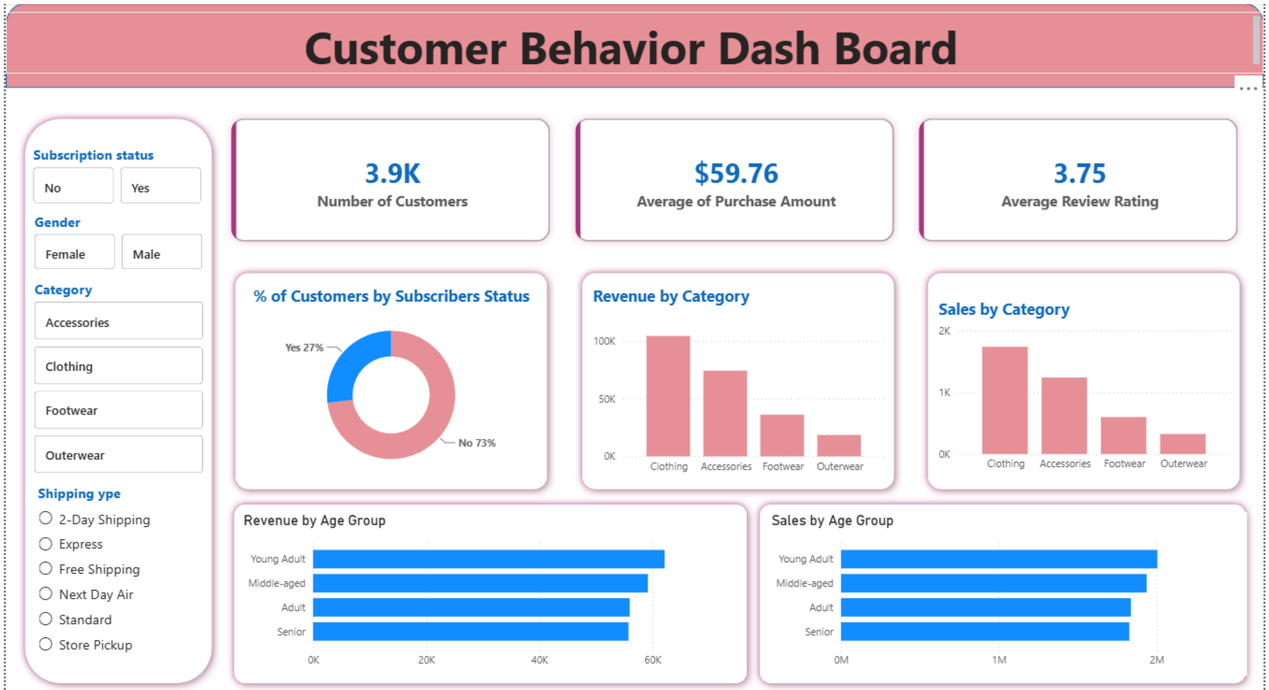
10. **Revenue by Age Group —** Calculated the total revenue contribution of each age group.

	revenue numeric	age_group text
1	55978	Adult
2	55763	Senior
3	62143	Young Adult
4	59197	Middle-aged

Total rows: 4 Query complete 00:00:00.306

5. Dashboard in Power BI

An interactive dashboard was built in **Power BI** to present the insights visually.



6. Business Recommendations

- **Encourage More Subscriptions —** Highlight special perks to attract more customers into the subscription program.
- **Reward Loyal Customers —** Introduce a loyalty program that motivates repeat buyers to become long-term, loyal customers.
- **Reassess Discount Strategy —** Fine-tune discount offers to drive sales without cutting too deep into profit margins.
- **Strengthen Product Marketing —** Feature high-rated and top-selling items more prominently in promotional campaigns.
- **Focus Marketing Efforts —** Direct marketing resources toward age groups that generate the most revenue and customers who prefer express shipping.